Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/12/2009
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Project Title:	Radiation Effects on Bone Tissue and Cells in Reduced Gravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI formerly at Clemson University when NSBRI Postdoctoral Fellow Feb 2008-Oct 2010 (Ed., 12/18/2014)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-02 Postdoctoral Fellowships
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2010 (from 1/31/2011) per NSBRI (8/2010) NOTE: End date changed to 1/31/2011 (from 1/31/2010) per NSBRI (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bateman, Ted (MENTOR/Clemson University)		
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Task Description:

POSTDOCTORAL FELLOWSHIP
The spaceflight environment presents numerous challenges to astronauts' bone health, including the bone-debilitating effects of micro/reduced gravity. A reduction in bone volume and strength among astronauts represents a serious, mission-critical biomedical problem. Preliminary data and published accounts indicate that exposure to doses as low as 1 Gy of multiple types of ionizing radiation results in trabecular bone loss. However, it is unclear how bone already compromised by microgravity or partial gravity will be affected by exposure to space radiation. Our intent is to enhance the understanding of the effects of radiation on bone health, particularly within a reduced gravity environment. Our ultimate goal is to develop appropriate countermeasures for bone loss in astronauts during exploratory missions. The proposed Aims will analyze the effects of modeled space radiation using scenarios applicable for Lunar Outpost missions.

Specific Aim 1: Examine the response of bone to a combination of spaceflight-associated challenges, including radiation exposure together with reduced limb bone loading. HYPOTHESIS: Proton or Fe radiation combined with unloading will induce more severe impacts on bone quantity and strength than unloading alone. Approach Aim 1: Skeletally mature female C57BL/6 mice, a strain sensitive to disuse-associated osteoporosis, were exposed to a 1 Gy dose or protons at Loma Linda Medical Center followed by a 4-week period of hindlimb suspension. Main Findings Aim 1: The combination of radiation and hindlimb suspension resulted in greater bone loss and deterioration of trabecular microarchitecture than the two challenges individually. Although both skeletal challenges induced atrophy, bone loss due to suspension alone was substantially greater than irradiation alone.

Specific Aim 2: Examine the mechanistic causes of changes in bone at the cellular and tissue level following the individual and combined effects of radiation and unloading on mouse hindlimbs. HYPOTHESIS: An early, inflammatory-type response following irradiation results in increased resorption via activation of osteoclasts. Approach Aim 2: Thirteen-week old C57BL/6 mice received whole body radiation with 2 Gy X-rays. X rays were chosen for these initial studies, as the relative biological effect of X-rays and protons are similar and bone loss from a 1 Gy dose has been shown to be similar (-13-15%) for both types at one month (preliminary data). Main Findings Aim 2: Radiation appears to increase osteoclast number and activity as an early, acute response to irradiation leading to early atrophy. Circulating markers of bone resorption are increased by 24 hours of exposure, with significantly increased osteoclast numbers present along trabeculae by 3 days. Trabecular bone loss is significant and substantial by Day 7 after exposure and occurs at multiple skeletal sites (proximal tibia; distal femur; fifth lumbar vertebrae). Bone quantity remains lower than control at Day 14 and 21. However, these changes in bone can completely be prevented by application of a commercially available osteoporosis medication (bisphosphonate) that prevents osteoclast activity.

These findings largely support the two hypotheses of the original proposal. Bone loss after exposure to multiple skeletal challenges appears additive relative to the two challenges individually. From Aim 2, radiation exposure resulted in a very early increase in osteoclast activity and number, with appreciable bone loss occurring by week 1. A large component of Aim 2 was to describe if the inflammatory response can lead to increased osteoclast activity early, as inflammation can induce bone loss and can occur within irradiated tissue. Thus, the findings from the first year have driven this research to focus on the following aspect of Aim 2: describe what cellular and molecular mechanism can lead to an increase in osteoclast activity early after exposure. Additionally, countermeasures (e.g., antiinflammatory and antiresorptive agents) will be employed in an attempt to both prevent radiation induced bone loss as well as to provide more information as to the nature of the increase in osteoclast activity. The research for the second year of NSBRI support will describe any role that early inflammation will play in this acute activation of osteoclasts. We will use primarily single limb irradiation to reduce the role of systemic confounding factors, such as changes in circulating estrogen if the gonads are irradiated. We will also focus on factors associated with inflammation. In particular, we will examine the response in bone after proinflammatory cytokine knock-out mice are exposed to radiation (e.g., tumor necrosis factor-alpha (TNFa), interleukin (IL)-1, and IL-6). These cytokines can activate osteoclasts. We will examine the role for transforming growth factor-beta (TGFb) in radiation-induced bone loss: this cytokine is often produced within irradiated tissues. This research will also examine if TNFa binding protein therapy or TGFb blockade with a type 1 receptor kinase inhibitor will block radiation-induced bone loss in this setting. Finally, we will continue to explore bisphosphonates as a countermeasure for radiation-induced bone loss.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The following Aim for this project supported by NSBRI has direct clinical relevance: Aim 2) Examine the mechanistic causes of changes in bone at the cellular and tissue level following irradiation and identify if bone loss can occur as an acute, early response; and test preventative countermeasures to prevent any early response that can lead to increased osteoclast activity. The notion that radiation can increase osteoclast activity early after exposure leading to bone loss is a novel cause for radiation-induced osteoporosis. A radiation-induced increase in osteoclast function represents a potential target to prevent late-observed bone loss and fractures among cancer survivors. Findings from the first year of this project suggest that: 1) radiation appears to increase osteoclast activity and number in an animal model during the first week after exposure, leading to substantial loss of bone by Day 7; and 2) the bone loss that occurs during the first week and any other evidence of deterioration can be prevented by applying a bisphosphonate as antiresorptive countermeasure. Bone loss is entirely prevented within the tibia, femur, and fifth lumbar vertebrae of mice by administering risedronate.
Radiation-induced bone damage is a concern for radiation oncologists. Improvements in cancer treatment and diagnosis have led to increased long-term cancer survivorship. For example, of the more than 200,000 men who are diagnosed with prostate cancer each year, the 10 year survival rate approaches 90%. However, this growing population of long-term survivors are at risk of developing deleterious effects in normal tissues resulting from the use of therapeutic radiation. Skeletal complications, particularly fractures, are a recognized late-radiation induced effect. Fractures of the hip, ribs, clavicle, and humerus have documented in patients receiving targeted radiotherapy for pelvic and breast tumors. These bones absorb radiation during cancer treatment. The increased incidence of hip (particularly the femoral neck) fractures in postmenopausal women after receiving pelvic cancer treatment is substantial and can have severe negative impacts on a patient's mobility, independence, and survival.

Bone atrophy and fractures following direct exposure to radiation is thought to occur as a late response resulting from damage to osteoblasts and vascularity within marrow and throughout osteons. Data suggest radiation can impair bone formation after exposure by killing or damaging osteoblasts. The contribution of late vascular damage to skeletal complications is less clear. However, late radiation-induced bone atrophy has been observed in clinical and animal studies.

	The effects of ionizing radiation on osteoclast activity are poorly documented, yet some evidence suggests an early stimulation of active bone resorption after exposure (as opposed to just reduced formation) could contribute to the etiology of radiation-induced bone damage. The possibility exists that an early, radiation-induced increase in osteoclast activity can contribute to late-observed atrophy. If this is true, atrophy of bone that is irradiated during the treatment of tumors as well as subsequent fractures of these structures may be reduced or prevented by early blockage of osteoclast activity using antiresorptive countermeasures, such as a bisphosphonate like risedronate.
Task Progress:	Overall, substantial progress was made for both Aim 1 and Aim 2 during the course of the first year. The Aims are to identify how bone will respond to multiple skeletal challenges associated with spaceflight, namely exposure to ionizing radiation and reduced loading in microgravity. Overall, radiation combined with hindlimb unloading results in greater loss of bone than radiation or hindlimb suspension alone. The two challenges appear additive. Radiation also appears to induce bone loss as an acute, early response. However, this bone loss and any evidence of deterioration can be prevented by applying an antiresorptive countermeasure for osteoporosis. Specific Aim 1: Examine the response of bone to a combination of spaceflight-associated challenges, specifically radiation exposure with reduced limb bone loading. Progress: Skeletally mature female C57BL/6 mice, a strain sensitive to disuse-associated osteoporosis, were exposed to a 1 Gy dose or protons at Loma Linda Medical Center, followed by a 4-week period of hindlimb suspension. Both of the skeletal challenges resulted in loss of bone volume fraction and deterioration of skeletal elements. The combination of radiation and hindlimb suspension resulted in greater bone loss and deterioration of trabecular microarchitecture than the two challenges individually. These changes appeared additive: relative reduction in bone volume fraction within the tibia after irradiation in the normally loaded mice (-15%) was very similar as the suspended mice (-17%) when compared with non-irradiated normally loaded and suspended mice, respectively. Specific Aim 2: Examine the mechanistic causes of changes in bone at the cellular and tissue level following irradiation and identify if bone loss can occur as an acute, early response. Test preventative countermeasures (antiresorptive; anti-inflammatory; antifibrotic) to prevent any early response that can lead to increased osteoclast activity. Progress: Radiation appears to increase osteoclast number and activity as an early, acute response to irradiation leading to early atrophy. C57BL/6 mice received whole body radiation with 2 Gy X-rays, as the relative biological effect of X-rays and protons are similar and bone loss from a 1 Gy dose has been shown to be similar (-13-15%) for both types at one month (preliminary data). Circulating serum markers of resorption (TRAP5b; +50%) are increased by 24 hours, which remains elevated by Day 3 (+14%) and Day 7 (+21%). Increased osteoclast numbers were present along trabeculae by Day 3 (+80%) and Day 7 (+218%). Surface covered by osteoclasts was also greater (+210%) at Day 3 after exposure. Trabecular bone loss is significant and substantial by Day 7 at multiple skeletal sites (proximal tibia; distal femur; fifth lumbar vertebrae). Deterioration bone was completely prevented by administering an antiresorptive osteoporosis countermeasure (risedronate).
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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Awards	Willey JS. "Radiation Research Society SIT Travel Award, Radiation Research Society Annual Meeting, Boston, Massachusetts, Sept 20-24, 2008." Sep-2008